

Federated Ontology Architecture

for

DoD Live-Virtual-Constructive (LVC) Integration

- Integrating Domain-Driven Federated Ontologies,
- AI Agent-Powered Semantic Mediation,
- Deterministic Circuit Breakers,
- Human-AI Teaming Frameworks

*Proposed Opportunities for Enhancing Trust, Context, & Integration in DoD M&S.
This is a consolidation and distillation of numerous documents and disciplines
integrated (with the help of AI) into a cohesive document.*

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1. Overview

The Department of Defense's Live-Virtual-Constructive (LVC) training ecosystem faces three interlocking challenges that no single framework can address in isolation. **First**, a semantic interoperability challenge: three distinct simulation environments with incompatible data models must function as a coherent operational picture. **Second**, a safety and trust challenge: as AI-mediated integration becomes more capable, the risk of model collapse, semantic drift, and unchecked autonomous action increases. **Third**, a human performance challenge: better automation paradoxically degrades the human oversight that makes the system trustworthy.

This unified architecture that addresses all three challenges through the integration of four complementary frameworks:

- **Federated Ontology Architecture:** Component-based ontology design using Domain-Driven Design (DDD) principles to define bounded contexts around each semantic domain, with AI agent-powered dynamic mapping services that mediate meaning across environment boundaries.
- **DEIX Consistency Management Model:** A reconceptualization of the LVC relationship where the Virtual component serves not as a coupled mirror but as an independent Consistency Manager—a distinct system responsible for judging the fidelity of Live observations against Constructive requirements.
- **Deterministic Circuit Breaker Framework:** Safety-critical boundary mechanisms that govern the conditional coupling between environments, preventing model collapse through Ground Truth authority, Recursive Loop Sanitization, and Semantic Guardrails encoding physical impossibilities.
- **Human-AI Teaming Framework:** Systematic engineering of the human dimension—trust calibration, cognitive workload management, competency maintenance, and production pressure detection—ensuring that human operators remain capable of effective oversight as automation improves.

The architecture treats LVC integration not as a protocol translation problem, nor as a pure AI automation problem, but as a socio-technical system engineering challenge where **semantic fidelity**, **operational safety**, and **human performance** must be co-designed to achieve trustworthy training outcomes at scale.

2. Problem Space: Three Interlocking Failure Modes

2.1 Semantic Interoperability Failure

Current LVC integration relies on protocol-level bridging between DIS and HLA, with LVC-IA providing gateway services that translate Protocol Data Units to Object Model interactions. While this enables entity-level connectivity, it introduces critical semantic gaps:

- **Semantic Drift:** In a live training setup, an M1A2 Abrams tank includes detailed information like its maintenance condition, crew skill level, and ammo status. But when that data is converted to standard simulation formats, most of those details get lost or oversimplified. So even though the tank appears across different simulation systems, its full real-world meaning doesn't carry over.

- **Combinatorial Mapping Explosion:** Each new simulation system requires point-to-point mappings to every other system, creating $O(N^2)$ complexity. Every time you add a new simulation system, you have to manually connect it to all the existing ones. As the number of systems grows, the number of these connections grows much faster — like a snowball effect. Because of that, even adding just one new simulation can take months of setup and testing before everything works together..
- **Context Collapse:** Different systems each use their own way of modeling things like terrain, weather, electronic warfare, and logistics. Since there's no shared standard or common language to connect these models, the systems don't naturally understand or line up with each other.
- **Temporal Inconsistency:** Live systems operate in real time, virtual systems may use compressed or dilated time, and constructive systems often use event-driven time advancement. No semantic model currently reconciles these temporal paradigms.

2.2 Safety and Trust Failure: The Coupling Problem

Companies like Dassault Systèmes advertise “coupled” Digital Twins, where a real system and its virtual model constantly share data and influence each other in real time. In the Department of Defense's Live, Virtual, and Constructive (LVC) training environment—where live equipment, virtual simulations, and computer-generated forces interact—this kind of tight two-way connection can be risky. If bad data, a cyberattack, or a model error affects the virtual twin, it could feed back into the real system and cause serious failures in equipment or training safety.:

- **Model Collapse:** When a Virtual Twin—basically a digital version of a real system—is too closely linked to the physical system, and there are no safety limits in place, it can start to “learn” from its own mistakes. Small errors in data or sensor noise get treated as real signals, and over time, those false lessons build up. The result is that the digital model stops accurately reflecting the real equipment—it drifts further away from reality.
- In the Live, Virtual, and Constructive (LVC) environment, this drift shows up as constructive entities (computer-generated forces or simulated units) starting to behave based on bad simulation data instead of real-world doctrine or tactics. That means the training loses realism, and decisions made on those faulty simulations can reinforce the wrong lessons.
- **Loss of Objectivity:** In a Live, Virtual, and Constructive (LVC) setup, the Live component refers to real systems—actual vehicles, weapons, sensors, and people operating in the physical world. The Virtual component simulates those systems in software to test scenarios, predict outcomes, or provide training safely and cheaply.
- When the Virtual and Live systems are too tightly linked, meaning the simulation directly mirrors and influences the live system in real time, the Virtual model can no longer act as an independent reference. Its job is to detect when the Live system drifts from expected behavior, experiences faults, or shows signs of tampering—but if they share the same data loop, any error or attack propagates instantly through both.
- In that case, the Virtual model can't offer an objective check on what's happening—**it's judging itself with the same potentially corrupted inputs**. That's why the phrase “the judge becomes the defendant” fits: the system meant to provide oversight becomes part of the problem it's supposed to detect.
- **“Super-Hubris” Assertions:** Without physical constraints encoded in the semantic model, the system can accept or generate impossible states—a 50-foot wall appearing in 2 seconds, a turbine spinning faster than physically possible without vibration—**that corrupt training fidelity**.

- **Cyber Vulnerability:** When real and virtual systems are tightly connected, they share data and control signals directly. This setup creates new cyber risks because an attacker could use the virtual system as a backdoor to reach the live one—or the other way around. Without an independent layer to check or verify the data moving between them, there's no way to tell if information or commands have been tampered with. That means **a cyberattack in the virtual environment could spill over and cause real-world damage or compromise mission safety.**

2.3 Human Performance Failure: The Automation Paradox

The most insidious failure mode is the one that emerges precisely as the technology improves. Research across aviation, healthcare, industrial control, and financial systems has documented a consistent pattern: better automation leads to worse human oversight, which leads to worse overall system performance.

In LVC systems, as AI tools get better at handling data mapping and integration, engineers start trusting them too much. When the AI “seems accurate”, people stop double-checking its work, engineers start **rubber-stamping semantic validations**, their own technical skills fade, and they're less prepared to step in when something goes wrong. This pattern has shown up in major failures like the Air France 447 crash, the Three Mile Island meltdown, and Knight Capital's trading disaster—technology worked so well that it made humans less capable of handling rare faults.

The risk can actually be measured. When AI is only about 65% accurate, humans stay alert and carefully review its output, keeping overall system reliability around 95%. But when AI accuracy climbs to 92%, people relax and stop verifying, and total reliability falls to around 85%. In other words, the smarter the AI appears, the worse the whole system can perform. This isn't just theory—**it's been consistently observed across different industries, making it a predictable human-technology failure pattern.**

3. The Unified System Model: System 1, 2, & 3

The architecture starts by rethinking how Live, Virtual, and Constructive (LVC) systems relate to one another, using the ASELCM — Agile Systems Engineering Life Cycle Management — pattern. Instead of viewing LVC as three simulation layers that merely need technical bridges, this approach defines them as three distinct but cooperating systems, each with a unique purpose: Live systems represent real-world operations and constraints; Virtual systems simulate controlled, interactive environments; and Constructive systems generate AI-driven or model-based scenarios for scaling complexity.

By applying agile methodology, development no longer follows rigid, sequential phases. Instead, engineers can iteratively design, test, and refine connections between LVC systems—improving fidelity and resilience with each sprint. Agile ceremonies (like short feedback loops and cross-functional reviews) keep software and hardware teams aligned, ensuring rapid adaptation to new data, doctrine, or mission requirements.

Lifecycle management ensures that every stage—from design through sustainment—is traceable and governed. This means updates to a simulation model, ontology, or integration interface are versioned and auditable. As LVC systems evolve, lifecycle traceability prevents drift between Live truth data and its Virtual or Constructive counterparts.

Semantic engineering provides the shared meaning layer that enables these systems to communicate reliably. Using ontologies and knowledge graphs, each environment can interpret data in context rather than just syntax. For example, a “mobility kill” in a Live feed carries the same definition and effect in the Constructive model, preventing misinterpretation across environments.

Finally, prompt engineering—a concept borrowed from AI—extends into systems engineering by defining structured, machine-interpretable requests or “prompts” that guide autonomous agents, analytics pipelines, or simulation controllers to perform context-aware actions. Engineers can craft prompts that, for instance, trigger scenario adjustments or semantic validations automatically, making LVC orchestration more intelligent and adaptive.

Combined, these elements turn LVC integration from a static, connector-heavy architecture into a living, knowledge-driven ecosystem—one that continuously learns, adapts, and maintains human oversight while scaling across technical domains.

Rather than treating Live, Virtual, and Constructive as three equivalent simulation environments to be bridged, we define them as three distinct systems with fundamentally different roles:

System	LVC Mapping	Role	Ontological Function
System 1	Live	The Engineered System—real entities operating in real environments with real instrumentation	Source of Ground Truth. Sensor data, GPS tracks, MILES engagements. Noisy, intermittent, but authoritative.
System 2	Virtual	The Consistency Manager—a distinct system responsible for judging the fidelity between Live observations and Constructive predictions	Not a mirror but a judge. Manages the Delta between intent and reality. Holds the Deterministic Veto.
System 3	Constructive	The Innovation/Wargaming Ecosystem—simulated entities and scenarios for planning, analysis, and capability exploration	Source of Intent. Doctrinal models, force-on-force wargames, course-of-action analysis. Idealized but valuable.

This new way of thinking changes the architecture in a big way. The Virtual component isn’t just a “coupled copy” of the Live system—it acts as an independent agent with a clear job: Consistency Management. It constantly compares the observed state of the Live system (System 1) with the predicted state from the Constructive system (System 3). If the difference—or Delta—goes beyond a defined Tolerance, the connection between the systems should be cut to prevent error or corruption. The Virtual Twin serves as the **judge measuring system fidelity**—and the judge cannot also be the defendant.

This setup also gives the Virtual component a cybersecurity function, known as the Digital Ghost pattern. Here, the Virtual Twin runs alongside the Live system, looking for unusual behavior that traditional IT or OT security might miss. If the Live system’s behavior breaks the rules of physics built into the Virtual model, that deviation could signal a hardware fault or a cyberattack. In that case, the Twin can step in—taking over control temporarily with a

trusted Virtual controller, performing what’s called the Neutralization Role.

4. Federated Ontology Architecture

4.1 Architectural Principles

The federated ontology is governed by seven foundational principles that integrate Domain Driven Design (DDD), semantic web standards, and trustworthy engineering (new discipline):

Principle	Application to LVC
Bounded Contexts	Each ontology component has clear ownership boundaries aligned to a domain. No component defines concepts owned by another. The Virtual Twin’s Consistency Management role is itself a bounded context.
Context Mapping	Explicit, formalized mappings between bounded contexts replace implicit assumptions. Mapping types include Shared Kernel (common upper ontology), Customer-Supplier (producer/consumer), and Anti-Corruption Layer (translation boundaries).
Ubiquitous Language	Each bounded context defines a precise vocabulary. The term “engagement” means one thing in the Weapons Effects ontology and another in the Training Assessment ontology—both are formally defined with no ambiguity.
Open-World Assumption	Missing data is explicitly unknown, not assumed. A constructive entity’s missing maintenance data is not “fully operational”—it is unknown, triggering appropriate handling by the Consistency Manager.
Composability	Ontology components assemble into mission-specific configurations. A maritime exercise composes different components than an urban operations scenario, but both share core entity and event semantics.
Versioned Evolution	Components evolve independently with semantic versioning. Breaking changes to inter-context contracts require explicit migration paths and AI-driven impact analysis.
Trust by Design	Trustworthiness is a measurable property, not an assertion. Every component, mapping, and agent output carries quantified trust metrics including provenance, confidence calibration, and Mean Time Between Hallucinations.

4.2 Four-Tier Component Model

4.2.1 Tier 1: Upper Ontology (Shared Kernel) with Trust Metrology

The shared kernel provides foundational concepts that all components depend upon. Based on the integration of the DEIX and Human-AI Teaming frameworks, Tier 1 now includes four components rather than the three in a purely structural model:

- **LVC-Core:** Defines Entity, Event, Environment, Agent, TemporalFrame, SpatialReference, and Provenance as top-level classes. Aligned to DOLCE foundational patterns. Includes the System 1/2/3 role taxonomy that distinguishes Live Ground Truth, Virtual Consistency Management, and Constructive Innovation roles.
- **LVC-Measurement:** Canonical units, coordinate reference systems, and measurement uncertainty models. Resolves unit mismatches across simulation environments and provides the quantitative foundation for Delta/Tolerance calculations.
- **LVC-Provenance:** Tracks the origin, transformation history, and confidence level of every assertion in the federation. Extended to include Recursive Loop Sanitization flags that mark data associated with Circuit Breaker trips for exclusion from agent learning.
- **LVC-Trust (New):** Elevates trustworthiness from metadata annotation to first-class ontological concern. Defines Trust Metrology including Mean Time Between Hallucinations (MTBH), Value-Drift Coefficient, confidence calibration curves, and Semantic Fidelity Index. Every entity, mapping, and agent output in the federation carries a Trust Profile grounded in this component.

This model aligns with the INCOSE Model-Based Capabilities Matrix (MBCM) pattern and Domain-Driven Design principles:

- **Level 1 Domains (Bounded Contexts):** Engineering disciplines—Systems Engineering, Software Engineering, Data Engineering, Knowledge Engineering, AI/ML Engineering, etc. These are the organizational units that own semantic territory, define ubiquitous language, and maintain independent governance authority.
- **Level 2 Sub-Domains:** Specializations within each discipline—e.g., under Knowledge Engineering: Ontology Engineering, Taxonomy Management, Semantic Governance, Knowledge Graph Operations. These refine the bounded context without breaking its boundary.
- **Shared Services/Assets:** Military subject matter artifacts (Entity Taxonomy, Terrain, C2, Engagement, Logistics, ISR, Communications, Training Assessment)—these are cross-cutting semantic assets that multiple engineering discipline domains produce, consume, validate, and govern. They are not domains themselves; they are the work products.

This distinction is critical because it determines where governance authority lives. An Entity Taxonomy is not a domain—it is an asset produced by Knowledge Engineering, consumed by Systems Engineering, validated by Data Engineering, and deployed by AI/ML Engineering. The bounded context boundaries must follow the organizational accountability, not the subject matter.

4.2.2: Level 1- Domain Ontologies (Bounded Contexts)

The federated ontology retains its four-tier architecture but redefines what constitutes a “domain” at each level. The critical change: Tier 2 bounded contexts are drawn around

engineering disciplines, not subject matter content. Subject matter is modeled as shared semantic assets that flow across discipline boundaries through governed interfaces.

Each bounded context corresponds to an engineering discipline with clear organizational ownership, defined governance authority, and independent evolvability. The ubiquitous language within each context reflects the discipline's terminology, not the military subject matter it operates on.

Level 1 Domain	Bounded Context Scope	Level 2 Sub-Domains
Systems Engineering	Requirements management, architecture definition, system integration, verification & validation, interface control across the LVC ecosystem	Requirements Engineering, Architecture & Design, Integration & Test, Interface Management, V&V, Configuration Management
Software Engineering	Development, testing, deployment, and maintenance of simulation software, gateway code, and integration middleware	Application Development, DevSecOps, Platform Engineering, API Management, Quality Assurance, Release Management
Data Engineering	Data pipelines, integration patterns, quality management, storage architectures, and data distribution across LVC environments	Data Pipeline Engineering, Data Quality Management, Data Integration, Data Storage & Distribution, Metadata Management, Data Lineage
Knowledge Engineering	Ontology design, taxonomy governance, semantic mapping, knowledge graph operations, and semantic consistency across the federation	Ontology Engineering, Taxonomy Management, Semantic Governance, Knowledge Graph Operations, Semantic Mapping & Alignment, Vocabulary Management
AI/ML Engineering	Model development, training, deployment, monitoring, and governance of AI agents that mediate semantic translation and mapping	Model Development & Training, MLOps & Deployment, Agent Architecture, Inference Optimization, AI Governance, Bias & Drift Detection
Simulation & Modeling Engineering	Design, calibration, and validation of simulation models across Live instrumentation, Virtual simulators, and Constructive wargames	Live Instrumentation, Virtual Simulation, Constructive Modeling, M&S V&V, Scenario Design, Terrain & Environment Modeling
Cybersecurity Engineering	Security architecture, threat modeling, anomaly detection, and resilience engineering across LVC attack surfaces including the Digital Ghost pattern	Security Architecture, Threat Modeling, Anomaly Detection, Penetration Testing, Incident Response, Digital Ghost Operations
Human Factors Engineering	Human-machine interface design, cognitive engineering, trust calibration, workload management, and competency maintenance for LVC operators	HMI Design, Cognitive Load Engineering, Trust Calibration, Competency Maintenance, Performance Monitoring, Training Design

Each Level 1 domain owns its ubiquitous language. When Knowledge Engineering says “mapping,” it means a semantic alignment between ontology concepts. When Data Engineering says “mapping,” it means a data transformation between schemas. When AI/ML Engineering says “mapping,” it means a learned correspondence between feature spaces. All three are formally defined within their bounded contexts and related through explicit inter-context contracts.

The military are classified as shared semantic assets—work products that are produced, consumed, validated, and governed across multiple engineering discipline domains. Each asset has a primary steward (the discipline with primary governance accountability) and multiple contributing and consuming disciplines.

4.2.3: Level 2 - Shared Semantic Assets (Cross-Domain Services)

The military subject matter areas that were previously modeled as domains are now correctly classified as shared semantic assets—work products that are produced, consumed, validated, and governed across multiple engineering discipline domains. Each asset has a primary steward (the discipline with primary governance accountability) and multiple contributing and consuming disciplines.

Semantic Asset	Primary Steward	Contributing Disciplines	Key Concepts
Entity Taxonomy	Knowledge Engineering (ontological structure) + Simulation & Modeling Engineering (entity fidelity)	Systems Eng (requirements), Data Eng (entity data pipelines), AI/ML Eng (entity classification models)	Platform, Weapon System, Personnel, Unit, Force Structure, Capability, Readiness State
Terrain & Environment	Simulation & Modeling Engineering (terrain models, weather) + Data Engineering (geospatial pipelines)	Knowledge Eng (terrain ontology), Cybersecurity Eng (environmental spoofing detection)	TerrainFeature, WeatherState, Visibility, Trafficability, EM Environment
Command & Control	Systems Engineering (C2 architecture) + Knowledge Engineering (C2 semantics)	Software Eng (C2 system integration), Human Factors Eng (operator interfaces)	Order, MissionPhase, TaskOrganization, DecisionPoint, ROE, BattleRhythm
Engagement & Effects	Simulation & Modeling Engineering (weapons models, BDA) + Knowledge Engineering (engagement semantics)	AI/ML Eng (automated BDA), Data Eng (engagement data fusion)	Engagement, Munition, DamageState, BDA, Attrition
Logistics & Sustainment	Systems Engineering (logistics architecture) + Data Engineering (supply chain data)	Knowledge Eng (logistics ontology), Simulation & Modeling Eng (logistics models)	SupplyClass, MaintenanceAction, CasualtyReport, SupplyRate, LogisticsNode

ISR & Intelligence	AI/ML Engineering (sensor fusion, target ID) + Simulation & Modeling Engineering (sensor models)	Data Eng (intelligence pipelines), Cybersecurity Eng (counter-ISR), Knowledge Eng (intel ontology)	Sensor, Detection, Track, IntelligenceProduct, TargetList, CollectionPlan
Communications & Networks	Software Engineering (network infrastructure) + Cybersecurity Engineering (COMSEC/spectrum)	Systems Eng (comms architecture), Simulation & Modeling Eng (network simulation)	CommNetwork, FrequencyAllocation, Bandwidth, CyberEffect, EMCON
Training Assessment	Human Factors Engineering (assessment methodology) + Knowledge Engineering (assessment ontology)	AI/ML Eng (automated assessment), Data Eng (AAR data pipelines)	TrainingObjective, Metric, Assessment, ObserverNote, LessonLearned
Consistency Mgmt	Knowledge Engineering (consistency semantics) + AI/ML Engineering (agent-based monitoring)	All disciplines (consumers of consistency state)	Delta, Tolerance, ConsistencyState, FidelityJudgment, DriftEvent, CircuitBreakerTrip
Cyber Resilience	Cybersecurity Engineering (threat detection) + Simulation & Modeling Engineering (physics reference models)	AI/ML Eng (anomaly detection models), Knowledge Eng (attack ontology)	AnomalySignature, PhysicsConstraint, NeutralizationEvent, IntegrityState, AttackVector

Knowledge Engineering owns the ontological structure, Simulation & Modeling Engineering owns the fidelity validation, Data Engineering owns the data pipelines, and Systems Engineering owns the requirements—each within their bounded context, with explicit inter-context contracts governing the interfaces.

4.2.4 Inter-Context Mapping Patterns

With engineering disciplines as bounded contexts, the DDD context mapping patterns take on precise meaning for how disciplines collaborate on shared semantic assets:

DDD Pattern	LVC Application	Example
Shared Kernel	The Tier 1 Upper Ontology that all disciplines depend upon. Changes require consensus from all bounded contexts.	LVC-Core Entity class is owned jointly. Any change to the definition of Entity requires sign-off from Systems, Knowledge, Simulation, and AI/ML Engineering.
Customer-Supplier	One discipline produces semantic assets that another consumes. The supplier commits to interface stability; the customer accepts the supplier's model.	Knowledge Engineering (supplier) produces the Entity Taxonomy ontology. Simulation & Modeling Engineering (customer) consumes it for entity initialization. KE commits to a stable taxonomy API.
Anti-Corruption Layer	A discipline builds a translation boundary to protect its internal model from external semantic contamination.	AI/ML Engineering wraps the Knowledge Engineering ontology in an ACL that translates OWL concepts into feature vectors for model training, insulating the ML pipeline from ontology schema changes.
Conformist	A discipline adopts another's model wholesale because the cost of translation exceeds the benefit of independence.	Data Engineering conforms to Knowledge Engineering's semantic mapping specifications for data pipeline transformations, using KE's vocabulary directly rather than maintaining a separate mapping model.
Published Language	A shared, well-documented interchange format that multiple disciplines agree to use for cross-boundary communication.	The DIS-Ontology and HLA-Ontology in Tier 4 serve as Published Languages—formal semantic models of existing standards that all disciplines can reference.
Open Host Service	A discipline exposes its capabilities through a standardized service interface that any other discipline can consume.	Knowledge Engineering exposes SPARQL endpoints per ontology component. AI/ML Engineering exposes agent APIs for translation, alignment, and reconciliation. Both are Open Host Services.

4.2.5 Tier 3: Environment Adaptation Ontologies (Anti-Corruption Layers)

These are not owned by a single discipline—they are collaborative products of Simulation & Modeling Engineering (fidelity), Data Engineering (data transformation), and Knowledge Engineering (semantic alignment):

- **Live-Adaptation:** Maps domain concepts to instrumentation system data (MILES/AESS gear, GPS tracks, RFID logistics tags). Handles the reality that live data is noisy, intermittent, and often requires inference. Includes explicit uncertainty models that propagate through to the Consistency Management bounded context.
- **Virtual-Adaptation:** Maps domain concepts to simulator-specific representations (CCTT, aviation simulators, dismounted soldier trainers). Incorporates the Consistency Manager's Delta/Tolerance calculations for each adaptation.
- **Constructive-Adaptation:** Maps domain concepts to wargame engine internal models (OneSAF, JICM, STORM). Handles aggregated vs. entity-level representation differences and explicitly models the gap between idealized constructive behavior and real-world physics.

4.2.6 Tier 4: Standards Integration Ontologies

These ontologies are categorized as a Published Language pattern in Domain Specific Details (DDD) terms. These ontologies are formal semantic models of existing DoD standards that all engineering discipline domains reference enabling backward-compatible interoperability:

- **DIS-Ontology:** Formalizes DIS PDU structures as ontological classes with semantic relationships to domain concepts. Enables semantic-level understanding of DIS traffic rather than purely syntactic parsing. Includes semantic enrichment of Entity State PDUs through the Entity Taxonomy, dead reckoning interpretation through Terrain & Environment context, and Fire/Detonation PDU chains through the Engagement ontology.
- **HLA-Ontology:** Maps HLA FOMs, SOMs, and RTI interaction models to the federated ontology. Formally encodes RPR-FOM 2.0 semantics for automatic alignment. Enables FOM/SOM semantic grounding through Alignment Agents during pre-exercise preparation.
- **LVC-IA-Ontology:** Models the LVC-IA gateway architecture, translation rules, and integration patterns as first-class ontological entities. Replaces LVC-IA's current syntactic translation approach with AI-mediated semantic mediation grounded in the Consistency Management bounded context.
- **JLCCTC-Ontology:** Captures exercise architecture, stimulation interfaces, and mission command system integration patterns. Formalizes the semantic contract between constructive simulation outputs and C2 system inputs, ensuring appropriate uncertainty, classification, and source attribution. Supports echelon bridging and multi-national interoperability through federated ontology extensions.

5. Deterministic Circuit Breaker Architecture

The Circuit Breaker is the safety-critical mechanism that governs the conditional coupling between LVC environments. It operationalizes the principle that coupling is not a permanent state but a conditional privilege, revocable when trust conditions are violated.

5.1 The Ground Truth Authority Hierarchy

The Circuit Breaker enforces a non-negotiable authority hierarchy that prevents model collapse and preserves the objectivity of the Consistency Manager:

Priority	Authority	Scope	Override Condition
1 (High)	Human Trustee	Ultimate Circuit Breaker. Can sever all coupling, override any automated decision, and recalibrate context.	None—the Human Trustee cannot be overridden by any automated system.
2	Live Sensor Physics	Deterministic Veto. Sensor-based physical measurements hold total authority over AI probabilistic suggestions.	Human Trustee only. No AI agent or constructive model can override validated sensor data.
3	Semantic Guardrails	Impossible State Detection. Formalized physical constraints encoded in the ontology that define what cannot happen.	Human Trustee can acknowledge known exceptions (e.g., training scenario deliberate impossibilities).
4	Consistency Manager	Delta/Tolerance judgments. The Virtual Twin's assessment of fidelity between Live and Constructive states.	Human Trustee or Sensor Physics can override. AI agents cannot self-override.
5 (Low)	AI Agent Suggestion	Mapping proposals, alignment recommendations, reconciliation suggestions from the agent layer.	Any higher authority. All suggestions require validation before becoming active mappings.

5.2 Circuit Breaker Operations

5.2.1 Delta/Tolerance Monitoring

The Consistency Manager continuously evaluates the Delta between System 1's observed state (Live sensor data) and System 3's predicted state (Constructive model output). Each domain ontology defines Tolerance thresholds appropriate to its concepts:

- **Entity Taxonomy:** Position Delta tolerance of 50m for ground vehicles, 200m for aircraft. Attribute state Delta tolerance varies by attribute criticality (weapons status has tighter tolerance than paint scheme).
- **Engagement & Effects:** Zero tolerance for impossible engagements (e.g., weapons employment exceeding platform capability envelope). Near-zero tolerance for damage state inconsistency.
- **Terrain & Environment:** Tolerance scaled to temporal resolution—weather state Delta at 15-minute intervals, terrain feature Delta at exercise initialization only.
- **Logistics & Sustainment:** Tolerance scaled to consumption rates—fuel state Delta at 5% per reporting period, ammunition state at per-round accuracy for critical munitions.

5.2.2 Trip Conditions and Response

When a Delta exceeds its Tolerance, the Circuit Breaker trips. The response depends on the severity and domain:

- **Warning (Delta 1.0-1.5x Tolerance):** Flag raised to integration engineer. AI agents increase monitoring frequency. Mapping confidence scores degraded. Human Trustee notified but not required to act immediately.
- **Trip (Delta >1.5x Tolerance):** Coupling severed for the affected entity/domain. Data flow continues but is quarantined for manual review. Reconciliation Agent invoked to propose resolution. Human Trustee required to approve re-coupling.
- **Emergency Halt (Semantic Guardrail Violation):** Immediate halt of all coupling for the affected system. Impossible State flagged across federation. Human Trustee required to investigate and explicitly re-authorize. Digital Ghost assessment triggered for potential cyber compromise.

5.2.3 Recursive Loop Sanitization

To prevent the system from learning from its own errors, all data associated with a Circuit Breaker trip is flagged in the LVC-Provenance component with a Sanitization marker. This marker:

- Excludes the data from AI agent training corpora, preventing model collapse
- Preserves the data in an audit trail for post-exercise forensic analysis
- Triggers a review queue entry for the AI Mapping Curator to determine root cause
- Increments the MTBH counter in the LVC-Trust component, degrading the trust profile of the affected mapping or agent

5.3 The Digital Ghost: Security Through Physics

The Digital Ghost extends the Circuit Breaker from a safety mechanism to a security mechanism. The Virtual Twin's high-fidelity physics models serve as a reference against which Live system behavior is validated:

- **Anomaly Detection:** If the Live system reports behavior that violates the physics encoded in the Virtual Twin's models—a turbine spinning faster than possible without vibration, a vehicle accelerating beyond its documented capability—the Digital Ghost flags this as a potential cyber attack or sensor fault.
- **Attack Classification:** The Cyber Resilience ontology classifies anomalies by attack vector, severity, and confidence. The Digital Ghost differentiates between sensor spoofing, data injection, model manipulation, and physical system compromise.
- **Neutralization:** If the Live system is confirmed compromised, the Digital Ghost can estimate correct signals from its physics models and temporarily replace the Live controller with a validated Virtual one. This is the Neutralization Role—the Virtual Twin becomes the source of truth until the Live system is restored.

6. AI Agent Architecture with Trust Engineering

The agent architecture extends the original four-class model to incorporate the Consistency Management role, Digital Ghost capability, and human performance monitoring. Five agent classes operate across the federation:

Agent Class	Function	Trust Engineering	Human Interaction
Translation Agent	Executes real-time semantic translations between bounded contexts using established mapping rules	Carries calibrated confidence scores with component breakdown. Flags overconfident mappings where historical accuracy < stated confidence.	Surfaces confidence explainability to engineers: string similarity, semantic similarity, structural alignment, historical precedent, domain usage scores.
Alignment Agent	Discovers potential mappings between ontology components by analyzing structural and semantic similarity	Proposes mappings with Trust Profiles. Cannot self-approve. All proposals enter human validation pipeline.	Provides evidence, concerns, and validation checklists. Detects when engineer is over-trusting at the proposed confidence level.
Reconciliation Agent	Detects and resolves conflicts when entities receive contradictory updates from different environments	Applies Ground Truth Authority Hierarchy. Cannot override sensor physics. Logs all resolutions with provenance.	Escalates to Human Trustee when confidence is below threshold. Manages Conflict Resolution Queue with prioritization.
Fidelity Agent	Monitors semantic loss during translations and flags degraded mappings that may impact training outcomes	Maintains MTBH and Value-Drift Coefficient for all active mappings. Triggers Circuit Breaker warnings.	Feeds data to post-exercise AAR. Correlates mapping fidelity with training assessment outcomes.
Validation Agent (New)	Implements the Digital Ghost pattern: physics-based anomaly detection against Live system behavior	Operates from independent physics models. Cannot be influenced by Live data feedback loops. Sanitized learning only.	Reports anomalies with attack classification and confidence. Recommends Neutralization only—Human Trustee authorizes.

6.1 Confidence Explainability

Drawing from the Human-AI Teaming Framework, every agent output includes a structured confidence explanation that enables informed human judgment rather than blind trust:

- **Component Scores:** Each confidence score is decomposed into weighted components: string similarity (25%), semantic similarity (30%), structural alignment (20%), historical precedent (15%), and domain usage (10%). Engineers see exactly where the score comes from.
- **Calibration Warning:** If historical accuracy at a given confidence level diverges from the stated confidence by more than 10%, the system warns the engineer. An 0.82 confidence score that historically achieves only 71% accuracy is flagged as overconfident, with a recommendation to treat it as 0.70.
- **Evidence and Concerns:** Positive evidence (exact matches, known synonyms, similar validated mappings) and negative concerns (stale schemas, multiple candidates, data type mismatches, confidence drift) are surfaced alongside every suggestion.
- **Validation Checklist:** Auto-generated, context-specific checklist that must be completed before acceptance. Critical items block the accept button until addressed, preventing perfunctory rubber-stamping.

7. Human-AI Teaming for LVC Operations

The Human-AI Teaming Framework extends trustworthy engineering principles to the people who supervise, validate, and intervene in the semantic mediation process. This is not an optional enhancement—it is a structural requirement. Without it, improved AI accuracy paradoxically degrades system reliability.

7.1 The Human Trustee Role

The Human Trustee is not a traditional “user” or “operator.” It is a formally defined governance role in the ontology with specific authorities and responsibilities:

- **Ultimate Circuit Breaker:** The Human Trustee holds the highest authority in the Ground Truth hierarchy. No automated system can override the Trustee’s decision to sever coupling, quarantine data, or halt operations.
- **Context Recalibration:** When the Digital Twin drifts, the Trustee re-calibrates context—redefining Tolerance thresholds, adjusting Semantic Guardrails for scenario-specific conditions, and resetting Trust Profiles after investigation.
- **Observability over Autonomy:** The Virtual Twin must prioritize explaining why it thinks the Live asset is performing a certain way over automatically correcting the Live asset. The Trustee makes the final call on corrective action.

7.2 Real-Time Performance Monitoring

Every validation decision by an integration engineer or Human Trustee is tracked with full context to detect degradation before it causes failures:

- **Fatigue Detection:** A composite score combining session duration (30%), time since break (30%), accuracy degradation (20%), and decision speed increase (20%). When the score exceeds 0.7, mandatory 15-minute break is enforced. Sessions exceeding six hours require termination.
- **Rushing Detection:** If decision speed exceeds 2.5x baseline, the system flags rushing behavior and recommends deceleration. In LVC operations, rushing during exercise execution can propagate semantic errors across all three environments simultaneously.
- **Accuracy Monitoring:** Rolling accuracy over the last 20 decisions is compared to baseline. A drop of more than 15% triggers a warning to review recent decisions, with automatic escalation if the trend continues.

7.3 Trust Calibration

For each confidence bucket (e.g., 0.80-0.85), the system tracks the engineer's acceptance rate against the actual accuracy rate:

- **Well-Calibrated (<5% gap):** No intervention needed. The engineer's trust level matches reality.
- **Slight Mis-calibration (5-15% gap):** Awareness message provided. Engineer receives periodic calibration feedback.
- **Over-Trust (>15% gap):** Active warning before each decision in the affected confidence range. Message shows the specific gap: "You accept 94% at this confidence level but actual accuracy is 71%. Please review carefully."
- **Under-Trust (>15% gap):** Notification that the AI is more reliable than the engineer's behavior suggests, reducing unnecessary validation overhead.

In LVC operations, trust mis-calibration is particularly dangerous because semantic mapping errors can cascade through all three environments and corrupt training outcomes for thousands of participants simultaneously.

7.4 Cognitive Workload Management

Task assignment is matched to cognitive capacity, not distributed randomly:

- **Capacity Assessment:** Each engineer's current cognitive state is assessed from session duration, break history, decision count, accuracy trends, and decision speed. The result is a capacity score from 0 (exhausted) to 1 (fresh).
- **Task Matching:** High-capacity engineers (>0.8) receive complex cross-domain mapping tasks to maintain engagement. Fatigued engineers (<0.5) receive simple single-domain validations. No engineer receives tasks that exceed their current capacity.
- **LVC-Specific Application:** During exercise execution, the highest-capacity engineers are assigned to cross-environment reconciliation tasks (the most consequential decisions), while routine single-environment validations are assigned to engineers with moderate capacity.

7.5 Competency Maintenance

To prevent skill atrophy, integration engineers perform periodic manual exercises without AI assistance:

- **Manual Mapping Exercises (Every 14 Days):** Engineer creates a semantic mapping from scratch, documents reasoning, and assigns confidence. Evaluated against AI suggestion and expert validation. Scores below 15% of baseline trigger remediation training.
- **Knowledge Assessments (Every 30 Days):** Domain knowledge quizzes covering ontology concepts, DIS/HLA semantics, and LVC integration principles.
- **Conflict Resolution Simulations (Every 21 Days):** Simulated cross-environment conflicts where the engineer must manually reconcile contradictory entity states—the exact skill needed when the Circuit Breaker trips.

This parallels the continuous validation applied to AI models. Capability is not assumed to persist—it is measured and maintained actively. An integration engineer who cannot manually perform a semantic mapping is a single point of failure, because when the AI fails, there is no fallback.

7.6 Production Pressure Detection

Organizational pressure to meet exercise timelines can erode safety culture. The system monitors for systemic indicators:

- **Individual Indicators:** Rushing (decisions $>2.5\times$ baseline speed), accuracy decline ($>10\%$), excessive alert overrides ($>20\%$), burnout risk (three or more consecutive 6+ hour sessions), off-hours work ($>30\%$ of sessions).
- **Team-Level Detection:** If more than 50% of the team shows rushing behavior, the cause is organizational (workload), not individual (carelessness). A composite pressure score above 0.7 triggers management alerts with specific interventions: halt non-critical work, reduce targets by 30-50%, enforce mandatory time off, conduct emergency staffing assessment.
- **LVC-Specific Risk:** Exercise preparation timelines create intense pressure to “just make it work”—accepting marginal mappings, overriding safety alerts, shortcutting validation procedures. These patterns are detectable and interruptible before they cause exercise failures.

8. Semantic Guardrails: The Ontology of Impossibility

A critical innovation in this architecture is the formal encoding of physical impossibilities in the ontology itself. Unlike SHACL constraints that validate data format, Semantic Guardrails validate physical reality. They define what cannot happen, creating a boundary beyond which no AI agent, simulation model, or data pipeline can assert state.

8.1 Classes of Impossible States

Category	Example
Physical Law Violations	A vehicle exceeds its documented maximum velocity on the specified terrain type
Temporal Impossibilities	A structural change occurs faster than physically possible (50-foot wall in 2 seconds)
Capability Violations	A weapon system engages a target beyond its maximum effective range or outside its engagement envelope
Conservation Violations	Ammunition expenditure exceeds basic load without resupply event in logistics ontology
Logical Contradictions	An entity is simultaneously reported as destroyed in one environment and operational in another

These guardrails serve dual purposes. For training fidelity, they prevent simulation artifacts from corrupting the exercise. For cybersecurity, they serve as the Digital Ghost’s detection mechanism—any state that violates physical law is either a system fault or an adversary action, and both require investigation.

9. Integrated Governance Framework

The governance framework integrates ontology governance, operational authority, and human performance oversight into a unified model:

Role	Responsibility
Human Trustee	Ultimate authority over coupling decisions, Circuit Breaker overrides, and context recalibration
Federation Steward	Integrity of the shared kernel and inter-context mapping contracts
Domain Owner	Content authority for a specific bounded context
Integration Engineer	Exercise-specific assembly, mapping validation, and intervention when Circuit Breaker trips
AI Mapping Curator	Review of AI-proposed mappings, agent confidence tuning, Recursive Loop Sanitization review
Performance Monitor (New)	Human performance oversight: fatigue detection, trust calibration, competency tracking, pressure detection

9.1 Governance Scope & Ownership

Governance Scope	Owner & Responsibilities
Entity Taxonomy	Knowledge Engineering owns ontological structure. Simulation & Modeling owns fidelity validation. Data Engineering owns data pipelines. Each within their bounded context.
Mapping Changes	Knowledge Engineering approves the semantic alignment. Data Engineering approves the pipeline implementation. AI/ML Engineering approves the agent configuration. Each approves within their authority.
Cross-environment translation fails	The AI/ML Engineering discipline (agent behavior) and Knowledge Engineering discipline (mapping correctness) share accountability, with the Integration Engineer as operational escalation point.
Semantic Guardrails	Knowledge Engineering defines the formal constraint rules. Simulation & Modeling Engineering validates them against physics models. Cybersecurity Engineering extends them for anomaly detection.
Tust calibration for human operators	Human Factors Engineering owns the calibration methodology, performance monitoring, and competency maintenance. AI/ML Engineering owns the confidence explainability that feeds calibration.

9.2 Change Management with Trust Impact Analysis

Ontology evolution follows semantic versioning with an integrated trust impact assessment:

Semantic Versioning: MAJOR.MINOR.PATCH. Major versions indicate breaking changes to bounded context interfaces; minor versions add concepts; patches correct definitions.

AI-Driven Impact Analysis: Before any change is approved, agents automatically analyze impact on downstream mappings, translation rules, Trust Profiles, and Semantic Guardrails.

Canary Deployment: Changes deployed in parallel with existing versions. AI agents monitor for semantic drift. New versions promoted only after Trust Metrology validates no degradation.

Human Trust Recalibration: When ontology changes alter the accuracy characteristics of AI mappings, trust calibration baselines are reset and engineers are notified that their historical calibration data may no longer apply.

10. Value Proposition

Dimension	Impact	Quantifiable Benefit	Risk Mitigated
Exercise Setup	AI-driven ontology composition and automated mapping replace manual integration engineering	60-80% reduction in integration preparation time	Schedule risk, integration labor cost
Semantic Fidelity	Meaning preservation with Trust Metrology and Semantic Guardrails across environment boundaries	Measurable fidelity improvement via MTBH and Value-Drift Coefficient	Training artifact corruption, unrealistic outcomes
Scalability	O(N) integration via ontology alignment replacing O(N ²) point-to-point mapping	Linear vs. quadratic scaling cost	Integration cost explosion with new federates
Safety	Circuit Breaker prevents model collapse, Semantic Guardrails prevent impossible states, Digital Ghost detects compromise	Zero tolerance for physics violations in production	Model collapse, cyber compromise, unsafe autonomous action
Human Reliability	Trust calibration, fatigue management, and competency maintenance prevent the Automation Paradox	System reliability at 99% vs. 85% without framework (measured at 92% AI accuracy)	Complacency, skill atrophy, rubber-stamping, \$10-20M preventable failures
JADC2 Alignment	Semantic data fabric with trust engineering directly supports JADC2 shared understanding vision	Training environment as proving ground for operational concepts	Training-operations gap in data architecture

12. Summary

The DoD's LVC ecosystem cannot achieve trustworthy integration through any single framework. Protocol bridging without semantic grounding produces syntactic connectivity without meaning. Semantic mediation without safety engineering produces capable systems that can fail catastrophically. AI-powered automation without human performance engineering produces a paradox where better technology creates worse outcomes.

This architecture integrates all four dimensions into a unified system:

- **Structural integrity** through federated ontologies with DDD bounded contexts, ensuring every concept has clear ownership, well-defined interfaces, and independent evolvability.
- **Behavioral integrity** through the System 1/2/3 model, where the Virtual Twin serves as an independent Consistency Manager rather than a coupled mirror, maintaining the objectivity required to judge fidelity.
- **Safety integrity** through Deterministic Circuit Breakers, Semantic Guardrails, Recursive Loop Sanitization, and the Digital Ghost pattern, ensuring the system cannot learn from its own errors, accept impossible states, or be silently compromised.
- **Human integrity** through trust calibration, cognitive workload management, competency maintenance, and production pressure detection, ensuring the people who supervise the system remain capable of effective oversight as the automation improves.

The result is an LVC integration architecture that is trustworthy not in principle but in practice—across all three environments, at scale, over time. The AI does not degrade because we monitor and maintain it. The humans do not degrade because we monitor and maintain them. The coupling between Live, Virtual, and Constructive is not permanent but conditional, governed by measurable trust and revocable when that trust is violated.

Every sophisticated automation system that neglected the human dimension has eventually experienced catastrophic failure. Every tightly coupled system that lacked safety boundaries has eventually experienced model collapse. Every integration architecture that relied on syntactic translation has eventually experienced semantic drift. This architecture learns from all three lessons simultaneously, creating a foundation for LVC training that the DoD can actually rely on—because the entire system, not just its components, is engineered for trustworthiness.